cp_12

1

Print Characters | 20/20

Solved

Easy

Given a string and `N` indices, where `N` is the length of the string. Write a program to print the character of the string present at each index of the given `N` indices.

Input

The first line of input contains a string.
The next `N` lines will contain an integer on each line.

Output

The output should be `N` lines containing character of the string present at the each index of the given `N` indices as shown in the sample output.

Explanation

For example, if the given string is `tarc` ,

- The length of the string is **4**.
- The given **4** indices are **3**, **1**, **2**, **0**.
- The character present at index **3** is **c**, at index **1** is **a**, at index **2** is **r** and at index **0** is **t**.

The output should be,

```
c
a
r
t
```

code:

```
string=input()
new_str=""
for i in range(1,len(string)+1):
    n=int(input())
    new_str=string[n]
    print(new_str)
```

2

Word Triangle | 15/15

Solved

Given a word, write a program that prints the letters of the word on `N` lines as shown in the sample output, where `N` is the length of the word.

Input

The input will be a single line containing a string.

Output

The output should be `N` lines containing letters of the word as shown in the sample output.

Explanation

For example, if the given word is `Rahul` ,

- The length of the word is **5**.
- The first line should contain the first letter **R**.
- The second line should contain the first two letters **Ra**.
- The third line will contain the first three letters **Rah**
- The fourth line will contain the first four letters **Rahu**
- The fifth line will contain all five letters **Rahul**

The output should be,

```
R
Ra
Rah
Rahu
Rahul
```

code:

```
string=input()
new_str=""
for i in range(len(string)):
    new_str+=string[i]
    print(new_str)
```

3

Square - 4 | 25/25

Solved

Easy

Given a number `N` , write a program to print a **Square** of `N` rows and `N` columns using numbers starting from **0**.

```
0 0 0 0 0
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
```

Note

There is a space after every number (`1`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows and `N` columns containing numbers starting from **0** as a **Square** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the numbers in **3** rows and **3** columns starting from **0** as shown in the given image.

code:

```
num=int(input())
for i in range(num):
    str_i=str(i)+" "
    row=str_i*num
    print(row)
```

4

Shuffle String | 20/20

Solved

Companies

Given a string and `N` indices, where `N` is the length of the string. Write a program to print the string after shuffling the characters in the order of the given `N` indices.

Input

The first line of input will contain a string.
The next N lines will contain an integer on each line.

Output

The output should be a single line containing the shuffled string.

Explanation

For example, if the given string is `goindc` ,

- The length of the string is **6**.
- The given **6** indices are **5**, **1**, **4**, **2**, **3** and **0**.
- The character present at index **5** is **c**, at index **1** is **o**, at index **4** is **d**, at index **2** is **i**, at index **3** is **n** and at index **0** is **g**.
- The shuffled string is **coding**.

The output should be **coding**.

Sample Input 1

```
goindc
5
1
4
2
3
0
```

Sample Output 1

```
coding
```

code:

```
string=input()
new_str=""
for i in range(len(string)):
    num=int(input())
    new_str+=string[num]
print(new_str)
```

5

Square - 5 | 25/25

Solved

Easy

Given a starting number `S` and a number `N`, write a program that prints a **Square** of `N` rows and `N` columns using numbers starting from `S`.

```
5 6 7
5 6 7
5 6 7
```

Note

There is a space after every number (`3`).

Input

The first line of input contains an integer representing the starting number `S`.
The second line of input contains an integer representing `N`.

Output

The output should be `N` rows and `N` columns containing numbers starting from `S` as a **Square** shown in the sample output.

Explanation

For example, if the given numbers are `S = 3` and `N = 4`,

- Print the numbers in **4** rows and **4** columns starting from **3** as shown in the given image.

code:

```
s=int(input())
n=int(input())
row=""

for i in range(n):
    row+=str(s+i)+" "


for i in range(n):
    print(row)
```

6

Count of Zeroes | 25/25

Solved

Easy

Given a number `N` , write a program to find the count of **0**'s in `N` .

Print **Count of zeroes is greater than three** if there are more than three **0**'s in the given number. Otherwise, print **Count of zeroes is not greater than three**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string. **Count of zeroes is greater than three** should be printed if there are more than three **0**'s in the given number. Otherwise, **Count of zeroes is not greater than three** should be printed.

Explanation

For example, if the given number is `N = 1030800` ,

- The digits in the given number **1030800** are **1, 0, 3, 0, 8, 0** and **0**.
- Number of zeroes in the given number `N` are **4**.
- There are more than **3** zeros (`0`) in the given number `N` . (`4 > 3`).

The output should be **Count of zeroes is greater than three**, as the given number has more than three **0**'s.

Sample Input 1

```
1030800
```

Sample Output 1

```
Count of zeroes is greater than three
```

code:

```
num=input()
count_of_0s=0
for each_char in num:
    condi=int(each_char)==0
    if condi:
        count_of_0s+=1
if count_of_0s>3:
    print("Count of zeroes is greater than three")
else:
    print("Count of zeroes is not greater than three")
```

7

Number of Even Digits | 25/25

Solved

Easy

Given a number `N` , write a program to find the count of even digits in `N` .

Print **Count of even digits is greater than two** if there are more than two even digits in the given number. Otherwise, print **Count of even digits is not greater than two**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string. **Count of even digits is greater than two** should be printed if there are more than two even digits in the given number. Otherwise, **Count of even digits is not greater than two** should be printed.

Explanation

For example, if the given number is `N = 2563408` ,

- The digits in the given number **2563408** are **2, 5, 6, 3, 4, 0** and **8**.
- The number of even digits in the given number is **5**. (25**6**3**4**0**8**).
- There are more than **2** even digits in the given number. (`5 > 2`).

The output should be **Count of even digits is greater than two**, as the given number has more than two even digits.

Sample Input 1

```
2563408
```

Sample Output 1

```
Count of even digits is greater than two
```

code:

```
num=input()
count_of_evens=0
for each_char in num:
    condition=int(each_char)%2==0
    if condition:
        count_of_evens+=1
if count_of_evens>2:
    print("Count of even digits is greater than two")
else:
    print("Count of even digits is not greater than two")
```

8

Prime Number | 35/35

Solved

Write a program to check whether the given number is a Prime number or not.

Note

A prime number is a number that has only two factors. The two factors of a prime number are 1 and itself.

Input

The input will be a single line containing an integer.

Output

The output should be a single line containing a string. `Prime Number` should be printed if the given number is a prime number. Otherwise `Not a Prime Number` should be printed.

Explanation

For example, if the given number is 5,

- The output should be `Prime Number` , as it is divisible only by itself and one.

For example, if the given number is 6,

- The output should be `Not a Prime Number` , as it is divisible by 1, 2, 3, and 6.

Sample Input 1

```
5
```

Sample Output 1

```
Prime Number
```

code:

```
num=int(input())
count=2

for i in range(3,num):
    if num%i==0:
        count+=1
if count>2:
    print("Not a Prime Number")
else:
    print("Prime Number")
```

9

Right Angled Triangle - 10 | 30/30

Solved

Easy

Given a number `N` , write a program to print a **Right Angled Triangle** of `N` rows using stars (`*`).

```
*
* * *
* * * * *
* * * * * * *
```

Note

- Each row should contain odd number of stars (`*`) starting from **1**.
- There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows, each row containing an odd number of stars (`*`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the stars (`*`) in **3** rows as shown in the given image.

code:

```
num=int(input())
for i in range(1,num*2):
    if i%2==1:
        row="* "*i
        print(row)
```

10

Inverted Right Angled Triangle - 7 | 30/30

So

Easy

Given a number `N` , write a program to print an **Inverted Right Angled Triangle** of `N` rows using stars (`*`).

```
* * *
* *
*
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

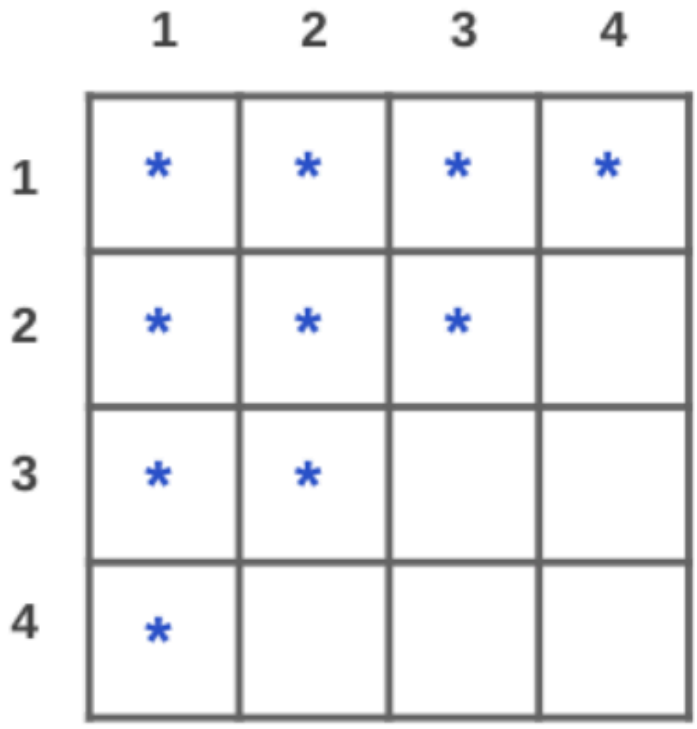
Output

The output should be `N` rows containing stars (`*`) as an **Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) in **4** rows as shown in the given image.



code:

```
n=int(input())
for i in range(n):
    row="* "*n
    n-=1
    print(row)
```

11

Hollow Square - 2 | 35/35

Solved

Easy

Given a number `N` , write a program that prints a **Hollow Square** of `N + 2` rows using pluses (`+`), hyphens (`-`), and pipes (`|`) . The first and last row contain pluses(`+`) and hyphens(`-`), and the middle rows contain pipes(`|`) as shown below.

```
+ - - - +
|       |
|       |
|       |
+ - - - +
```

Note

There is a space after every character.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N + 2` lines containing pluses (`+`), hyphens (`-`) and pipes (`|`) as a **Hollow Square** shown in the sampl output.

Explanation

For example, if the given number is `N = 5` ,

- The **Hollow Square** should contain **7** lines. (`5 + 2 = 7`)
- The first and last lines should contain **2** pluses (`+`) and **5** hyphens (`-`), as shown below.

```
+ - - - - - +
```

- The middle **5** lines should contain two pipes (`|`) with **10** spaces between them, as shown below.

```
|           |
|           |
|           |
|           |
|           |
```

- The final **Hollow Square** is as shown in the given image.

code:

```
num=int(input())
for i in range(num+2):
    if i==0 or i==(num+1):
        row="+ "+"- "*(num)+"+ "
        print(row)
```

```
    else:
        row="|  "+"   "*(num)+"|  "
        print(row)
```

12

Case Conversion - 3 | 30/30

Solved

Easy

Given a title case string, write a program to modifiy the string as given below.

- Add a hyphen(-) before each uppercase character excluding the first uppercase character.
- Convert all uppercase characters into lowercase characters.

Note

Title case is a writing style where the first letter of every word in a sentence is capitalized, except for certain words like articles and prepositions.

Input

The input will be a single line containing a string.

Output

The output should be a single line containing a string, as shown in the sample output.

Explanation

For example, if the given string is **"TitleCase"**,

- The uppercase characters in the given string **TitleCase** are **T** and **C**.
- The modified string after adding hyphen **(-)** before each uppercase character excluding the first uppercase character is **Title-Case**.
- The modified string after converting the uppercase characters **T, C** to the lower case characters **t, c** is **title-case**.

The output should be **"title-case"**.

Sample Input 1

```
TitleCase
```

Sample Output 1

```
title-case
```

code:

```
string=input()
new_str=string[0]+""
for i in range (1,len(string)):
```

```
    upper_case=string[i].upper()
    if string[i]==upper_case:
        new_str+=string[i].replace(string[i],("-"+string
[i]))
    else:
        new_str+=string[i]
lower=new_str.lower()
print(lower)
```

13

Numbers in Square Pattern - 2 | 25/25 ● Solved

Write a program to print numbers from 1 to N in each column forming a square pattern of N rows and N columns.

Input

The input will be a single line containing a positive integer (N).

Output

The output should be N rows containing a number N times in each row.
Note: There is a space after each number.

Explanation

For example, if the given number is 5, your code should print the following pattern:

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

Sample Input 1

```
5
```

Sample Output 1

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

code:

```
n=int(input())
for i in range(1,n+1):
```

```
    str_i=str(i)+" "
    row=str_i*n
    print(row)
```

14

Numbers in Rectangular Pattern - 1 | 30/30 ● Solved

Write a program to print numbers from 1 to N in each row forming a rectangular pattern of M rows and N columns.

Input

The first line of input will contain a positive integer (M).
The second line of input will contain a positive integer (N).

Output

The output should be M number of lines and N number of columns with the numbers from 1 to N in each row.
Note: There is a space after each number.

Explanation

For example, if the given numbers of rows are 2 and the number of columns is 3, your code should print the following pattern:

```
1 2 3
1 2 3
```

Sample Input 1

```
2
3
```

Sample Output 1

```
1 2 3
1 2 3
```

Sample Input 2

```
5
4
```

Sample Output 2

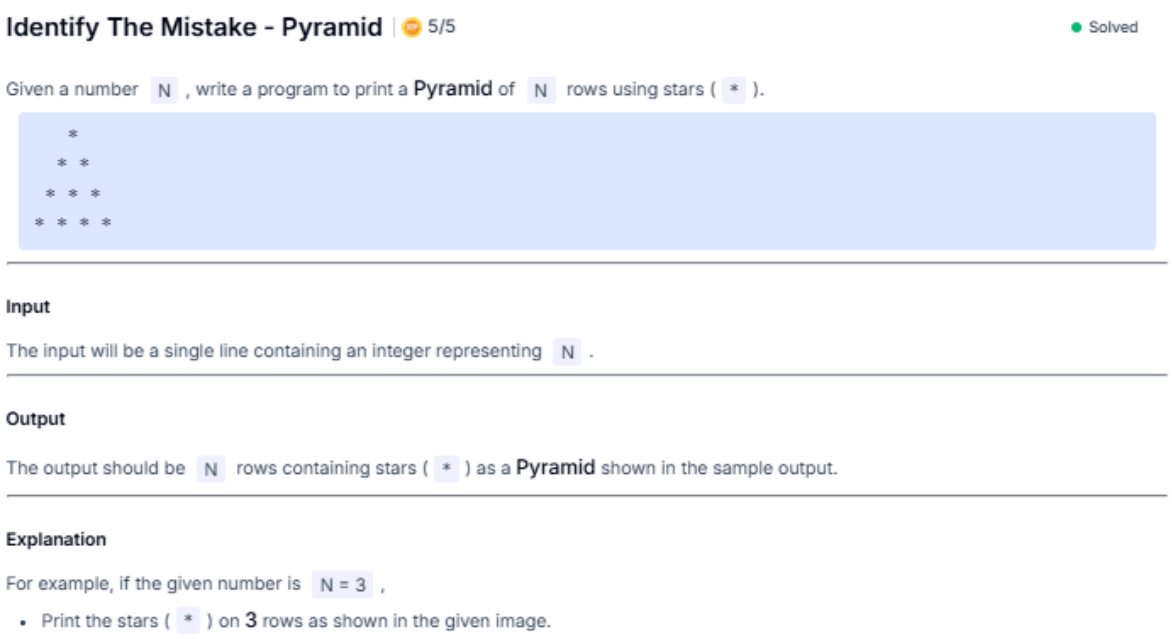
```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

code:

```
m=int(input())
n=int(input())
```

```
row=""
for i in range(n):
    row+=str(i+1)+" "
for i in range(m):
    print(row)
```

15



Identify The Mistake - Pyramid | 5/5 — Solved

Given a number `N` , write a program to print a **Pyramid** of `N` rows using stars (`*`).

```
   *
  * *
 * * *
* * * *
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the stars (`*`) on **3** rows as shown in the given image.

code:

```
n = int(input())

k = n
for i in range(1, n+1):
 spaces = " " * (k-1)
 stars = "* " * i
 print(spaces+stars)
 k = k - 1
```